Supplementary Material for *Large-scale traffic flow reconstruction from heterogeneous mobility sensors*

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This Supplementary Material documents the data layers, the simulation model, the route-choice variants, the detector correction and additional validation results of the main manuscript, and lists the commands that reproduce them.

S1 Data layers

Table S1: Data layers and their role in the reconstruction.

Layer	Content	Role	Period
Mobile-phone OD (TIM)	hourly device movements between census areas, aggregated to municipalities and provinces	demand	January 2020
Road graph	3,925 directed polylines, 2,548 nodes, turn restrictions, derived from OpenStreetMap	network	–
OSM road classes	motorway to unclassified, nearest ways within 30 m	route-choice parameters	extract of 26/10/2026
MTS loop detectors	15-min counts per station and direction	local correction; volume validation	Feb 2020, Jun–Jul 2021
Viasat GPS, link counts	hourly probes per matched link	profile validation	10–17/02/2020, 28/06–04/07/2021
Viasat GPS, raw traces	private-vehicle fixes in Forlì–Cesena	speeds and class usage for route choice	same periods

The OD tables are hourly. Their labels refer to the start of the hour in local time: the weekday OD profile of 10–14 February 2020 and the mean profile of the 90 detectors complete on those days have correlation 0.96 at zero lag, 0.94 with the OD profile shifted one hour earlier and 0.74 with it shifted one hour later. Demand is generated in four classes: observed source and destination (`src-dest`, municipalities, and `scr_dest_prov`, provinces), observed source and random destination (`src-rnd`), random source and observed destination (`rnd-dest`) and random source and destination (`rnd-rnd`). On a weekday, the random-endpoint classes account for 66% of the generated agents (`rnd-rnd` 33%, `src-rnd` 22%, `rnd-dest` 12%).

MTS counts of 2021 were obtained from the monthly open-data files of Regione Emilia-Romagna (dataset DS-1944, CC BY), summing all vehicle classes. The open-data time labels are in local time;

converted to UTC they reproduce the counts of the February 2020 archive used for 2020 (ratio of totals 0.995 on 12 February 2020).

S2 Simulation model

Agents move with the speed of their link, v_i, reduced linearly with density ρ_i below the critical density and set to a critical speed above it. Link speeds are read in km/h from the cartography and assigned by road class from GPS (main text, Table 1). Random-endpoint agents have a trip-time budget of 5,600 s; source–destination agents follow the best-path matrix of their type. When the turn-restricted best-path search finds no feasible next node for the cost of their type, the agent uses the shortest-distance path matrix; this happened for 50 next-node decisions in two of the fifteen simulated days. The simulator is deterministic (fixed random seed). A simulated day takes 9–12 minutes on an 8-core workstation running four days in parallel.

S3 Route-choice variants

Table S2 lists the variants simulated for 10 February 2020. Speeds alone (V1–V3) do not change the distribution of traffic over road classes, because two thirds of the agents follow transition probabilities rather than best paths. Transition weights by road class (V4–V6) raise the share of main roads and the agreement with GPS counts in the calibration area. The selected variant combines the usage cost of Eq. (1) with class transition weights. Correlations with the detectors of a single day (53 coils) are noisy and were not used for selection; the comparison over full campaigns is in the main text. Table S3 shows that route choice hardly changes the hourly profiles.

Table S2: Route-choice variants simulated for 10 February 2020. Calibration: Spearman correlation between simulated daily link volumes and GPS counts on Forlì–Cesena links. Detectors: Spearman correlation with the daily volumes of the coils outside Forlì–Cesena complete on that day (not used for selection). Main roads: share of simulated vehicle-kilometres on motorway, trunk and primary links.

Variant	GPS links	Calibration ρ_s	Detectors	Detector ρ_s	Main roads
Shortest distance, default speeds	814	0.474	53	0.198	0.39
Shortest distance, GPS speeds	814	0.478	53	0.190	0.38
Travel time, GPS speeds	814	0.487	53	0.167	0.38
Mixed distance/time, GPS speeds	814	0.484	53	0.175	0.37
Travel time + class transition weights	814	0.537	53	0.109	0.47
Travel time + damped class weights	814	0.532	53	0.105	0.42
Usage cost + class transition weights (selected)	814	0.542	53	0.176	0.48

Table S3: Median hourly-share MAE (pp) and hourly correlation on all usable GPS links with shortest-distance and GPS-informed route choice.

Campaign	Route choice	Links	Raw	Assimilated	Correlation
2020	shortest distance	821	1.46	1.40	0.81
2020	GPS-informed	821	1.52	1.41	0.81
2021	shortest distance	818	1.51	1.48	0.76
2021	GPS-informed	818	1.50	1.50	0.76

S4 Detector correction

The correction starts from the residual of Eq. (2) on each coil link and hour. The ratio $r_i(h)$ converts the coil count, a count of passages at a section, into the scale of the simulated unique-link count, which counts distinct agents present on the whole polyline during the hour. Downstream, the increment of a link is passed to each admissible outgoing link multiplied by that link's share of the simulated outgoing counts; upstream, the same rule is applied to incoming links. Each branch carries its own increment and its own path length. Propagation stops when the increment is below one vehicle, after three polylines along the path, at another coil link, at a forbidden turn or at a closed polyline. Every directed polyline of the raw simulation is kept in the output, and the hours of the observed table are aligned with the local-time hours of the simulation.

Table S4 summarizes the leave-one-detector-out test, Table S5 the dependence on the propagation depth and Table S6 the effect of the correction on the hourly profiles of GPS links by network distance from the nearest coil.

Table S4: Leave-one-detector-out test. Errors are $\sum |\tilde{u} - c| / \sum c$ in percent of the observed volume, pooled over coils or as the median over coils; reached: share of withheld coils whose link is modified by the propagation from other coils; improved/worse: share with lower/higher error than the raw reconstruction.

Year	Coils	Coil-days	Pooled raw	Pooled LOO	Median raw	Median LOO	Reached (%)	Improved (%)	Worse
2020	152	614	78.3	88.1	71.7	67.8	64	45	
2021	140	571	75.9	84.2	72.4	69.2	57	39	

Table S5: Change of the pooled error (percentage points of observed volume) on a fixed random 20% of the complete coils of each 2020 day when all other coils are assimilated with the given propagation depth. Raw error of the test coils: 75.5%. Negative values are increases of the error.

Depth (polylines)	Error reduction (pp)
1	-5.7
3	-13.6
5	-12.3

Table S6: Median hourly-share MAE (pp) of the raw and assimilated reconstruction on GPS links by network distance from the nearest coil (polylines).

Year	Distance	Links	Raw	Assimilated	Reduction
2020	0	55	1.26	0.87	0.40
2020	1	198	1.45	1.26	0.19
2020	2	262	1.43	1.37	0.06
2020	3	185	1.56	1.47	0.10
2020	4-5	110	2.05	2.05	0.00
2020	6+	11	2.09	2.09	0.00
2021	0	55	1.23	1.01	0.22
2021	1	198	1.38	1.38	0.00
2021	2	262	1.45	1.44	0.01
2021	3	184	1.58	1.59	-0.01
2021	4-5	108	2.01	2.01	0.00
2021	6+	11	2.01	2.01	0.00

S5 Additional validation results

Table S7 reports the profile error by number of probes per link. The error decreases with the number of probes for both the reconstruction and the MTS profile, as expected when part of the difference is sampling noise of the probe counts. Matched links with zero simulated or GPS total are excluded; absent link-hour records of the GPS extract are counted as zero.

Table S7: Hourly-share MAE (pp) by number of GPS probes per link and campaign: reconstruction, mean MTS profile and hourly correlation of the reconstruction (medians).

Campaign	Probes	Links	Reconstruction	MTS	Correlation
2020	1-49	119	3.76	3.71	0.41
2020	50-199	191	2.04	1.82	0.66
2020	200+	511	1.10	0.82	0.88
2021	1-49	124	3.94	3.84	0.35
2021	50-199	224	1.98	1.81	0.68
2021	200+	470	1.18	0.89	0.84

S6 Reproducibility

All results are produced by the scripts in `article1/analysis` of the code repository, run from its root:

1. `od_clock_check.py`: hour convention of the OD tables;
2. `carto_geojson.py`, `osm_road_class.py`, `gps_link_speeds.py`: geometry, road classes and GPS link speeds;
3. `routing_variants.py prepare|run|evaluate`: route-choice variants and selection;
4. `run_corrected_sims.py`: simulations of the validation days;
5. `build_coils_table.py`: 2021 detector table from the open-data files;
6. `reconstruct_and_validate.py assimilate|validate|loo`: correction, GPS validation, leave-one-out;
7. `volume_check.py`, `compare_routing.py`, `distance_to_detector.py`, `sensitivity_fixed.py`: volume tests, routing comparison, reach of the correction, depth sensitivity;
8. `make_figures.py`, `make_supp_tables.py`: figures and tables.

Proprietary inputs (mobile-phone OD tables, Viasat link counts and traces) are required for steps 1–6 and are not distributed.